

Name: _____

A Level Mathematics : Pure Mathematics

Topic 1 :Proof

Date:

Time:

Total marks available:

Total marks achieved: _____

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Questions

Q1.

Given that θ is measured in radians, prove, from first principles, that the derivative of $\sin \theta$ is $\cos \theta$

You may assume the formula for $\sin(A \pm B)$ and that as $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$

(5)

(Total for question = 5 marks)

Q2.

Complete the table below. The first one has been done for you.

For each statement you must state if it is always true, sometimes true or never true, giving a reason in each case.

Statement	Always True	Sometimes True	Never True	Reason
The quadratic equation $ax^2 + bx + c = 0$, ($a \neq 0$) has 2 real roots.		✓		It only has 2 real roots when $b^2 - 4ac > 0$. When $b^2 - 4ac = 0$ it has 1 real root and when $b^2 - 4ac < 0$ it has 0 real roots.
(i) When a real value of x is substituted into $x^2 - 6x + 10$ the result is positive.				
(2)				
(ii) If $ax > b$ then $x > \frac{b}{a}$				
(2)				
(iii) The difference between consecutive square numbers is odd.				
(2)				

(Total for question = 6 marks)

Q3.

The curve C has equation

$$y = 3x^4 - 8x^3 - 3$$

(a) Find (i) $\frac{dy}{dx}$

(ii) $\frac{d^2y}{dx^2}$

(3)

(b) Verify that C has a stationary point when $x = 2$

(2)

(c) Determine the nature of this stationary point, giving a reason for your answer.

(2)

(Total for question = 7 marks)

Q4.

(a) "If m and n are irrational numbers, where $m \neq n$, then mn is also irrational."

Disprove this statement by means of a counter example.

(2)

(b) (i) Sketch the graph of $y = |x| + 3$

(ii) Explain why $|x| + 3 \geq |x + 3|$ for all real values of x .

(3)

(Total for question = 5 marks)

Q5.

(a) Prove that

$$\tan \theta + \cot \theta \equiv 2 \operatorname{cosec} 2\theta, \quad \theta \neq \frac{n\pi}{2}, n \in \mathbb{Z}$$

(4)

(b) Hence explain why the equation

$$\tan \theta + \cot \theta = 1$$

does not have any real solutions.

(1)

(Total for question = 5 marks)

Q6.

Given

$$f(x) = e^x, \quad x \in \mathbb{R}$$

$$g(x) = 3 \ln x, \quad x > 0, x \in \mathbb{R}$$

(a) find an expression for $gf(x)$, simplifying your answer.

(2)

(b) Show that there is only one real value of x for which $gf(x) = fg(x)$

(3)

(Total for question = 5 marks)

Q7.

The speed of a small jet aircraft was measured every 5 seconds, starting from the time it turned onto a runway, until the time when it left the ground.

The results are given in the table below with the time in seconds and the speed in ms^{-1} .

Time (s)	0	5	10	15	20	25
Speed (ms^{-1})	2	5	10	18	28	42

Using all of this information,

(a) estimate the length of runway used by the jet to take off.

(3)

Given that the jet accelerated smoothly in these 25 seconds,

(b) explain whether your answer to part (a) is an underestimate or an overestimate of the length of runway used by the jet to take off.

(1)

(Total for question = 4 marks)

Q8.

(i) Prove that for all $n \in \mathbb{N}$, $n^2 + 2$ is not divisible by 4

(4)

(ii) "Given $x \in \mathbb{R}$, the value of $|3x - 28|$ is greater than or equal to the value of $(x - 9)$."

State, giving a reason, if the above statement is always true, sometimes true or never true.

(2)

(Total for question = 6 marks)

Q9.

Prove, using algebra, that

$$n(n^2 + 5)$$

is even for all $n \in \mathbb{N}$.

(Total for question = 4 marks)

Q10.

(i) Given that p and q are integers such that

pq is even

use algebra to prove by contradiction that at least one of p or q is even.

(3)

(ii) Given that x and y are integers such that

- $x < 0$
- $(x + y)^2 < 9x^2 + y^2$

show that $y > 4x$

(2)

(Total for question = 5 marks)

Q11.

A company decides to manufacture a soft drinks can with a capacity of 500 ml.

The company models the can in the shape of a right circular cylinder with radius r cm and height h cm.

In the model they assume that the can is made from a metal of negligible thickness.

(a) Prove that the total surface area, S cm², of the can is given by

$$S = 2\pi r^2 + \frac{1000}{r} \quad (3)$$

Given that r can vary,

(b) find the dimensions of a can that has minimum surface area.

(5)

(c) With reference to the shape of the can, suggest a reason why the company may choose not to manufacture a can with minimum surface area.

(1)

(Total for question = 9 marks)

Q12.

$$f(x) = -3x^3 + 8x^2 - 9x + 10, \quad x \in \mathbb{R}$$

(a) (i) Calculate $f(2)$

(ii) Write $f(x)$ as a product of two algebraic factors.

(3)

Using the answer to (a)(ii),

(b) prove that there are exactly two real solutions to the equation

$$-3y^6 + 8y^4 - 9y^2 + 10 = 0 \quad (2)$$

(c) deduce the number of real solutions, for $7\pi \leq \theta < 10\pi$, to the equation

$$3 \tan^3 \theta - 8 \tan^2 \theta + 9 \tan \theta - 10 = 0 \quad (1)$$

(Total for question = 6 marks)

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
	Use of $\frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta}$	B1	2.1
	Uses the compound angle identity for $\sin(A+B)$ with $A = \theta$, $B = h$ $\Rightarrow \sin(\theta+h) = \sin \theta \cos h + \cos \theta \sin h$	M1	1.1b
	Achieves $\frac{\sin(\theta+h) - \sin \theta}{h} = \frac{\sin \theta \cos h + \cos \theta \sin h - \sin \theta}{h}$	A1	1.1b
	$= \frac{\sin h}{h} \cos \theta + \left(\frac{\cos h - 1}{h} \right) \sin \theta$	M1	2.1
	Uses $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$ and $\frac{\cos h - 1}{h} \rightarrow 0$ Hence the $\lim_{h \rightarrow 0} \frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \cos \theta$ and the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos \theta$ *	A1*	2.5
(5 marks)			

Notes:	
B1:	States or implies that the gradient of the chord is $\frac{\sin(\theta+h) - \sin \theta}{h}$ or similar such as $\frac{\sin(\theta+\delta\theta) - \sin \theta}{\theta + \delta\theta - \theta}$ for a small h or $\delta\theta$
M1:	Uses the compound angle identity for $\sin(A+B)$ with $A = \theta$, $B = h$ or $\delta\theta$
A1:	Obtains $\frac{\sin \theta \cos h + \cos \theta \sin h - \sin \theta}{h}$ or equivalent
M1:	Writes their expression in terms of $\frac{\sin h}{h}$ and $\frac{\cos h - 1}{h}$
A1*:	Uses correct language to explain that $\frac{dy}{d\theta} = \cos \theta$
For this method they should use all of the given statements $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$, $\frac{\cos h - 1}{h} \rightarrow 0$ meaning that the $\lim_{h \rightarrow 0} \frac{\sin(\theta+h) - \sin \theta}{(\theta+h) - \theta} = \cos \theta$ and therefore the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos \theta$	

Question	Scheme	Marks	AOs
alt	Use of $\frac{\sin(\theta+h)-\sin\theta}{(\theta+h)-\theta}$	B1	2.1
	Sets $\frac{\sin(\theta+h)-\sin\theta}{(\theta+h)-\theta} = \frac{\sin\left(\theta+\frac{h}{2}+\frac{h}{2}\right)-\sin\left(\theta+\frac{h}{2}-\frac{h}{2}\right)}{h}$ and uses the compound angle identity for $\sin(A+B)$ and $\sin(A-B)$ with $A=\theta+\frac{h}{2}$, $B=\frac{h}{2}$	M1	1.1b
	Achieves $\frac{\sin(\theta+h)-\sin\theta}{h} = \frac{\left[\sin\left(\theta+\frac{h}{2}\right)\cos\left(\frac{h}{2}\right)+\cos\left(\theta+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)\right]-\left[\sin\left(\theta+\frac{h}{2}\right)\cos\left(\frac{h}{2}\right)-\cos\left(\theta+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)\right]}{h}$	A1	1.1b
	$= \frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \times \cos\left(\theta+\frac{h}{2}\right)$	M1	2.1
	Uses $h \rightarrow 0$, $\frac{h}{2} \rightarrow 0$ hence $\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \rightarrow 1$ and $\cos\left(\theta+\frac{h}{2}\right) \rightarrow \cos\theta$ Therefore the limit $\lim_{h \rightarrow 0} \frac{\sin(\theta+h)-\sin\theta}{(\theta+h)-\theta} = \cos\theta$ and the gradient of the chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos\theta$ *	A1*	2.5
(5 marks)			

Additional notes:

A1*: Uses correct language to explain that $\frac{dy}{d\theta} = \cos\theta$. For this method they should use the

(adapted) given statement $h \rightarrow 0$, $\frac{h}{2} \rightarrow 0$ hence $\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \rightarrow 1$ with $\cos\left(\theta+\frac{h}{2}\right) \rightarrow \cos\theta$

meaning that the limit $\lim_{h \rightarrow 0} \frac{\sin(\theta+h)-\sin\theta}{(\theta+h)-\theta} = \cos\theta$ and therefore the gradient of the

chord $\rightarrow$ gradient of the curve $\Rightarrow \frac{dy}{d\theta} = \cos\theta$

Q2.

Question	Scheme	Marks	AOs
(i)	$x^2 - 6x + 10 = (x-3)^2 + 1$	M1	2.1
	Deduces "always true" as $(x-3)^2 \geq 0 \Rightarrow (x-3)^2 + 1 \geq 1$ and so is always positive	A1	2.2a
		(2)	
(ii)	For an explanation that it need not (always) be true This could be if $a < 0$ then $ax > b \Rightarrow x < \frac{b}{a}$	M1	2.3
	States 'sometimes' and explains if $a > 0$ then $ax > b \Rightarrow x > \frac{b}{a}$ if $a < 0$ then $ax > b \Rightarrow x < \frac{b}{a}$	A1	2.4
		(2)	
(iii)	Difference $= (n+1)^2 - n^2 = 2n+1$	M1	3.1a
	Deduces "Always true" as $2n+1 = (\text{even} + 1) = \text{odd}$	A1	2.2a
		(2)	
(6 marks)			

Notes:

(i)

M1: Attempts to complete the square or any other valid reason. Allow for a graph of $y = x^2 - 6x + 10$ or an attempt to find the minimum by differentiation

A1: States always true with a valid reason for their method

(ii)

M1: For an explanation that it need not be true (sometimes). This could be if

$$a < 0 \text{ then } ax > b \Rightarrow x < \frac{b}{a} \text{ or simply } -3x > 6 \Rightarrow x < -2$$

A1: Correct statement (sometimes true) and explanation

(iii)

M1: Sets up the proof algebraically.

For example by attempting $(n+1)^2 - n^2 = 2n+1$ or $m^2 - n^2 = (m-n)(m+n)$ with $m = n+1$

A1: States always true with reason and proof

Accept a proof written in words. For example

If integers are consecutive, one is odd and one is even

When squared odd $\times$ odd = odd and even $\times$ even = even

The difference between odd and even is always odd, hence always true

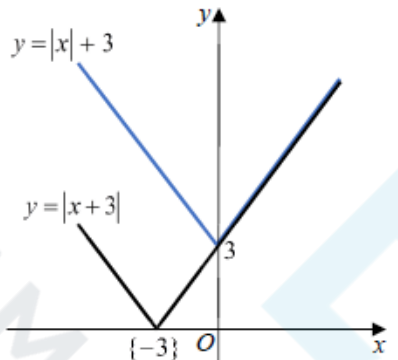
Score M1 for two of these lines and A1 for a good proof with all three lines or equivalent.

Q3.

Question	Scheme	Marks	AOs
(a)	(i) $\frac{dy}{dx} = 12x^3 - 24x^2$	M1 A1	1.1b 1.1b
	(ii) $\frac{d^2y}{dx^2} = 36x^2 - 48x$	A1ft	1.1b
		(3)	
(b)	Substitutes $x = 2$ into their $\frac{dy}{dx} = 12 \times 2^3 - 24 \times 2^2$	M1	1.1b
	Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point"	A1	2.1
		(2)	
(c)	Substitutes $x = 2$ into their $\frac{d^2y}{dx^2} = 36 \times 2^2 - 48 \times 2$	M1	1.1b
	$\frac{d^2y}{dx^2} = 48 > 0$ and states "hence the stationary point is a minimum"	A1ft	2.2a
		(2)	
(7 marks)			

Notes:
(a)(i) M1: Differentiates to a cubic form A1: $\frac{dy}{dx} = 12x^3 - 24x^2$ (a)(ii) A1ft: Achieves a correct $\frac{d^2y}{dx^2}$ for their $\frac{dy}{dx} = 36x^2 - 48x$
(b) M1: Substitutes $x = 2$ into their $\frac{dy}{dx}$ A1: Shows $\frac{dy}{dx} = 0$ and states "hence there is a stationary point" All aspects of the proof must be correct
(c) M1: Substitutes $x = 2$ into their $\frac{d^2y}{dx^2}$ Alternatively calculates the gradient of C either side of $x = 2$ A1ft: For a correct calculation, a valid reason and a correct conclusion. Follow through on an incorrect $\frac{d^2y}{dx^2}$

Q4.

Question	Scheme		Marks	AOs
	Statement: "If m and n are irrational numbers, where $m \neq n$, then mn is also irrational."			
(a)	E.g. $m = \sqrt{3}$, $n = \sqrt{12}$		M1	1.1b
	$\{mn = \} \quad (\sqrt{3})(\sqrt{12}) = 6$ $\Rightarrow$ statement untrue or 6 is not irrational or 6 is rational		A1	2.4
			(2)	
(b)(i), (ii) Way 1		V shaped graph {reasonably} symmetrical about the y-axis with vertical intercept (0, 3) or 3 stated or marked on the positive y-axis	B1	1.1b
		Superimposes the graph of $y = x + 3 $ on top of the graph of $y = x + 3$	M1	3.1a
	the graph of $y = x + 3$ is either the same or above the graph of $y = x + 3 $ {for corresponding values of x } or when $x \geq 0$, both graphs are equal (or the same) when $x < 0$, the graph of $y = x + 3$ is above the graph of $y = x + 3 $		A1	2.4
			(3)	
(b)(ii) Way 2	<u>Reason 1</u> When $x \geq 0$, $ x + 3 = x + 3 $	Any one of Reason 1 or Reason 2	M1	3.1a
	<u>Reason 2</u> When $x < 0$, $ x + 3 > x + 3 $	Both Reason 1 and Reason 2	A1	2.4
(5 marks)				

Notes for Question			
(a)			
M1:	States or uses any pair of <i>different</i> numbers that will disprove the statement. E.g. $\sqrt{3}$, $\sqrt{12}$; $\sqrt{2}$, $\sqrt{8}$; $\sqrt{5}$, $-\sqrt{5}$; $\frac{1}{\pi}$, 2π ; $3e$, $\frac{4}{5e}$;		
A1:	Uses correct reasoning to disprove the given statement, with a correct conclusion		
Note:	Writing $(3e)\left(\frac{4}{5e}\right) = \frac{12}{5} \Rightarrow$ untrue is sufficient for M1A1		
(b)(i)			
B1:	See scheme		
(b)(ii)			
M1:	For constructing a method of comparing $ x +3$ with $ x+3 $. See scheme.		
A1:	Explains fully why $ x +3 \geq x+3 $. See scheme.		
Note:	Do not allow either $x > 0$, $ x +3 \geq x+3 $ or $x \geq 0$, $ x +3 \geq x+3 $ as a valid reason		
Note:	$x = 0$ (or where necessary $x = -3$) need to be considered in their solutions for A1		
Note:	Do not allow an incorrect statement such as $x \leq 0$, $ x +3 > x+3 $ for A1		
(b)(ii)			
Note:	Allow M1A1 for $x > 0$, $ x +3 = x+3 $ and for $x \leq 0$, $ x +3 \geq x+3 $		
Note:	Allow M1 for any of x is positive, $x +3 = x+3$ x is negative, $x +3 > x+3$ $x > 0$, $x +3 = x+3$ $x \leq 0$, $x +3 \geq x+3$ $x > 0$, $x +3$ and $x+3$ are equal $x \geq 0$, $x +3$ and $x+3$ are equal - when $x \geq 0$, both graphs are equal - for positive values $x +3$ and $x+3$ are the same Condone for M1 $x \leq 0$, $x +3 > x+3$ $x < 0$, $x +3 \geq x+3$		
(b)(ii) Way 3	- For $x > 0$, $x +3 = x+3$ - For $-3 < x < 0$, as $x +3 > 3$ and $\{0 < x+3 < 3\}$, then $x +3 > x+3$ - For $x \leq -3$, as $x +3 = -x+3$ and $x+3 = -x-3$, then $x +3 > x+3$	M1	3.1a
		A1	2.4

Q5.

Question	Scheme	Marks	AOs
(a)	$\tan \theta + \cot \theta \equiv \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$	M1	2.1
	$\equiv \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$	A1	1.1b
	$\equiv \frac{1}{\frac{1}{2} \sin 2\theta}$	M1	2.1
	$\equiv 2 \operatorname{cosec} 2\theta \quad *$	A1*	1.1b
		(4)	
(b)	States $\tan \theta + \cot \theta = 1 \Rightarrow \sin 2\theta = 2$ AND no real solutions as $-1 \leq \sin 2\theta \leq 1$	B1	2.4
		(1)	
(5 marks)			

Notes:	
(a)	M1: Writes $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\cot \theta = \frac{\cos \theta}{\sin \theta}$ A1: Achieves a correct intermediate answer of $\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$ M1: Uses the double angle formula $\sin 2\theta = 2 \sin \theta \cos \theta$ A1*: Completes proof with no errors. This is a given answer. Note: There are many alternative methods. For example $\tan \theta + \cot \theta \equiv \tan \theta + \frac{1}{\tan \theta} \equiv \frac{\tan^2 \theta + 1}{\tan \theta} \equiv \frac{\sec^2 \theta}{\tan \theta} \equiv \frac{1}{\cos^2 \theta \times \frac{\sin \theta}{\cos \theta}} \equiv \frac{1}{\cos \theta \times \sin \theta} \text{ then as the}$ main scheme.
(b)	B1: Scored for sight of $\sin 2\theta = 2$ and a reason as to why this equation has no real solutions. Possible reasons could be $-1 \leq \sin 2\theta \leq 1$.....and therefore $\sin 2\theta \neq 2$ or $\sin 2\theta = 2 \Rightarrow 2\theta = \arcsin 2$ which has no answers as $-1 \leq \sin 2\theta \leq 1$

Q6.

Question	Scheme	Marks	AOs
(a)	$gf(x) = 3 \ln e^x$	M1	1.1b
	$= 3x, (x \in \mathbb{R})$	A1	1.1b
		(2)	
(b)	$gf(x) = fg(x) \Rightarrow 3x = x^3$	M1	1.1b
	$\Rightarrow x^3 - 3x = 0 \Rightarrow x =$	M1	1.1b
	$\Rightarrow x = (+)\sqrt{3}$ only as $\ln x$ is not defined at $x = 0$ and $-\sqrt{3}$	M1	2.2a
		(3)	
(5 marks)			
Notes:			
(a)			
M1: For applying the functions in the correct order			
A1: The simplest form is required so it must be $3x$ and not left in the form $3 \ln e^x$			
An answer of $3x$ with no working would score both marks			
(b)			
M1: Allow the candidates to score this mark if they have $e^{3 \ln x} =$ their $3x$			
M1: For solving their cubic in x and obtaining at least one solution.			
A1: For either stating that $x = \sqrt{3}$ only as $\ln x$ (or $3 \ln x$) is not defined at $x = 0$ and $-\sqrt{3}$			
or stating that $3x = x^3$ would have three answers, one positive one negative and one zero but $\ln x$ (or $3 \ln x$) is not defined for $x \leq 0$ so therefore there is only one (real) answer.			
Note: Student who mix up fg and gf can score full marks in part (b) as they have already been penalised in part (a)			

Q7.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$(\frac{1}{2} \times 5)(42 + 2 + 2 \times (5 + 10 + 18 + 28))$	M1	This mark is given for a method to use the trapezium rule as an approximation to the area under the curve
		M1	This mark is given for a correct terms used for the trapezium rule
	415 m	A1	This mark is given for a correct estimate of the length of the runway
(b)	An overestimate since the area of the five trapezia is greater than the area under the curve	B1	This mark is given for a valid explanation
(Total 4 marks)			

Q8.

Part	Working or answer an examiner might expect to see	Mark	Notes
(i)	For even numbers $n = 2m$, $n^2 + 2 = 4m^2 + 2$	M1	This mark is given for showing the case for all even numbers
	This is a multiple of 4 with 2 added, so cannot be divisible by 4	A1	This mark is given for a correct conclusion with a reason why $n^2 + 2$ is not divisible by 4 for all even numbers
	For odd numbers $n = 2m + 1$, $n^2 + 2 = (2m + 1)^2 + 2 = 4m^2 + 4m + 3$ $= 4(m^2 + m) + 3$	M1	This mark is given for showing the case for all odd numbers
	This is a multiple of 4 with 3 added, so cannot be divisible by 4 Hence, for all $n \in \mathbb{N}$, $n + 2$ is not divisible by 4	A1	This mark is given for a correct conclusion with a reason why $n^2 + 2$ is not divisible by 4 for all odd numbers and a full concluding statement that for all $n \in \mathbb{N}$, $n + 2$ is not divisible by 4
(ii)	For example, for $x = 9.4$ $ 3x - 28 = 0.2$ and $(x - 9) = 0.4$	M1	This mark is given for showing that the statement is not true for $9.25 < x < 9.5$
	The statement is sometimes true; For example, for $x = 12$ $ 3x - 28 = 8$ and $(x - 9) = 3$	A1	This mark is given for a correct statement and an example where the statement is true

Q9.

Question	Scheme	Marks	AOs
	$n(n^2 + 5)$		
	Attempts even or odd numbers Sets $n = 2k$ or $n = 2k \pm 1$ oe and attempts $n(n^2 + 5)$	M1	3.1a
	Achieves $2k(4k^2 + 5)$ (for $n = 2k$) and states "even" Or achieves $(2k+1)(4k^2 + 4k + 6) = 2(2k+1)(2k^2 + 2k + 3)$ (for $n = 2k+1$) and states "even" Or e.g. achieves $(2k-1)(4k^2 - 4k + 6) = 2(2k-1)(2k^2 - 2k + 3)$ (for $n = 2k-1$) and states "even"	A1	2.2a
	Attempts even and odd numbers Sets $n = 2k$ and $n = 2k \pm 1$ oe and attempts $n(n^2 + 5)$	dM1	2.1
	Achieves $2k(4k^2 + 5)$ (for $n = 2k$) and states "even" and achieves $(2k \pm 1)(4k^2 \pm 4k + 6) = 2(2k \pm 1)(2k^2 \pm 2k + 3)$ (for $n = 2k \pm 1$) and states "even" Correct work and states even for both WITH a final conclusion showing that true for all $n (\in \mathbb{N})$ or e.g. true for all even and odd numbers.	A1	2.4
		(4)	
(4 marks)			
Notes:			

M1: For the key step attempting to find $n(n^2 + 5)$ when $n = 2k$ or $n = 2k \pm 1$ or equivalent representation of odd or even e.g. $n = 2k + 2$ or $n = 2k \pm 7$
Condone the use of e.g. $n = 2n$ and $n = 2n \pm 1$

A1: Achieves $2k(4k^2 + 5)$ or e.g. $2(4k^3 + 5k)$ and deduces that this is even at the appropriate time.
Or achieves $(2k \pm 1)(4k^2 \pm 4k + 6) = 2(2k \pm 1)(2k^2 \pm 2k + 3)$ oe e.g. $2(4k^3 + 6k^2 + 8k + 3)$ and deduces that this is even.

Note that if the bracket is expanded to e.g. $8k^3 + 12k^2 + 16k + 6$ then stating "even" is insufficient – they would need to say e.g. even + even + even + even = even or equivalent

Note it is also acceptable to use a divisibility argument e.g. $\frac{8k^3 + 10k}{2} = 4k^3 + 5k$ so $8k^3 + 10k$ must be even.

There should be no errors in the algebra but allow e.g. invisible brackets if they are "recovered".

dM1: Attempts $n(n^2 + 5)$ when $n = 2k$ and $n = 2k \pm 1$ or equivalent representation of odd or even
e.g. $n = 2k + 2$ and $n = 2k \pm 7$

A1: Correct work and states even for both WITH a final conclusion e.g. so true for all $n (\in \mathbb{N})$.

There should be no errors in the algebra but allow e.g. invisible brackets if they are "recovered".

A "solution" via just logic.

E.g.

If n is odd, then $n(n^2 + 5)$ is $\text{odd} \times (\text{odd} + \text{odd}) = \text{odd} \times \text{even} = \text{even}$

If n is even, then $n(n^2 + 5)$ is $\text{even} \times (\text{even} + \text{odd}) = \text{even} \times \text{odd} = \text{even}$

Both cases must be considered to score any marks and scores SC 1010 if fully correct

OR

E.g. $n(n^2 + 5) = n^3 + 5n$

If n is odd, then n^3 is odd and $5n$ is odd, so $n^3 + 5n$ is $\text{odd} + \text{odd} = \text{even}$

If n is even, then n^3 is even and $5n$ is even, so $n^3 + 5n$ is $\text{even} + \text{even} = \text{even}$

Both cases must be considered to score any marks and scores SC 1010 if fully correct

A solution via contradiction.

M1 A1: There exists a number n such that $n(n^2 + 5)$ is odd, and so deduces that both n and $n^2 + 5$ are odd. Note that M1A0 is not possible via this method.

dM1: Sets $n^2 + 5 = 2k + 1$ (for some integer k) $\Rightarrow n^2 = 2k - 4 = 2(k - 2)$ which is even
Must use algebra here for this approach and not a "logic" argument.

A1: States that "this is a contradiction as if n^2 is even, then n is even" and then concludes so " $n(n^2 + 5)$ is even for all n ."

Attempts at proof by induction should be sent to review

Q10.

Question	Scheme	Marks	AOs
(i)	For setting up the contradiction: There exists integers p and q such that pq is even and both p and q are odd	B1	2.5
	For example, sets $p = 2m + 1$ and $q = 2n + 1$ and then attempts $pq = (2m + 1)(2n + 1) = \dots$	M1	1.1b
	Obtains $pq = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1$ $= 2(2mn + m + n) + 1$ States that this is odd, giving a contradiction so "if pq is even, then at least one of p and q is even" *	A1*	2.1
		(3)	
(ii)			
	$(x + y)^2 < 9x^2 + y^2 \Rightarrow 2xy < 8x^2$	M1	2.2a
	States that as $x < 0 \Rightarrow 2y > 8x$ $\Rightarrow y > 4x$ *	A1*	2.1
		(2)	
(5 marks)			
Notes:			

(i)

B1: For using the "correct"/ allowable language in setting up the contradiction.

Expect to see a minimum of

- "assume" or "let" or "there is " or other similar words
- " pq is even" and " p and q are (both) odd"

M1: Uses a correct algebraic form for p and q and attempting to multiply.

Allow any correct form so $p = 2n + 1$ and $q = 2m + 3$ would be fine to use

Different variables must be used for p and q , so $p = 2n + 1$ and $q = 2n - 1$ would be M0

A1*: Full argument .

This requires (1) a correct calculation for their pq

(2) a correct reason and conclusion that it is odd

E.g. $(2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1 = \text{odd}$

E.g. $(2m - 1)(2n + 1) = 4mn + 2m - 2n - 1 = \text{even} + \text{even} - \text{even} - 1 = \text{odd}$

and (3) a minimal statement implying that they have proven what was required which could be QED, proven etc

Note that B0 M1 A1 is possible

(ii)

M1: For multiplying out and cancelling terms before proceeding to a correct intermediate line such as $2xy < 8x^2$ o.e. such as $2x(4x - y) > 0$

A1*: Full and rigorous proof with reason shown as to why inequality reverses. The point at which it reverses must be correct and a correct reason given

See scheme

$$\text{Alt: } 2xy < 8x^2 \Rightarrow xy - 4x^2 < 0 \Rightarrow x(y - 4x) < 0$$

$$\text{as } x < 0, (y - 4x) > 0 \Rightarrow y > 4x \text{ scores M1 A1}$$

So, the following should be scored M1 A0 as line 3 is incorrect

$$2xy - 8x^2 < 0$$

$$\Rightarrow 2xy < 8x^2$$

$$\Rightarrow y < 4x$$

$$\Rightarrow y > 4x \text{ as } x < 0$$

There should be no incorrect lines in their proof

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Q11.

Question	Scheme	Marks	AOs
(a)	Sets $500 = \pi r^2 h$	B1	2.1
	Substitute $h = \frac{500}{\pi r^2}$ into $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \times \frac{500}{\pi r^2}$	M1	2.1
	Simplifies to reach given answer $S = 2\pi r^2 + \frac{1000}{r}$ *	A1*	1.1b
		(3)	
(b)	Differentiates S with both indices correct in $\frac{dS}{dr}$	M1	3.4
	$\frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}$	A1	1.1b
	Sets $\frac{dS}{dr} = 0$ and proceeds to $r^3 = k$, k is a constant	M1	2.1
	Radius = 4.30 cm	A1	1.1b
	Substitutes their $r = 4.30$ into $h = \frac{500}{\pi r^2} \Rightarrow$ Height = 8.60 cm	A1	1.1b
		(5)	
(c)	States a valid reason such as - The radius is too big for the size of our hands - If $r = 4.3$ cm and $h = 8.6$ cm the can is square in profile. All drinks cans are taller than they are wide - The radius is too big for us to drink from - They have different dimensions to other drinks cans and would be difficult to stack on shelves with other drinks cans	B1	3.2a
		(1)	
9 marks			

Notes:

(a)

B1: Uses the correct volume formula with $V=500$. Accept $500 = \pi r^2 h$

M1: Substitutes $h = \frac{500}{\pi r^2}$ or $rh = \frac{500}{\pi r}$ into $S = 2\pi r^2 + 2\pi r h$ to get S as a function of r

A1*: $S = 2\pi r^2 + \frac{1000}{r}$ Note that this is a given answer.

(b)

M1: Differentiates the given S to reach $\frac{dS}{dr} = Ar \pm Br^{-2}$

A1: $\frac{dS}{dr} = 4\pi r - \frac{1000}{r^2}$ or exact equivalent

M1: Sets $\frac{dS}{dr} = 0$ and proceeds to $r^3 = k$, k is a constant

A1: $R =$ awrt 4.30 cm

A1: $H =$ awrt 8.60 cm

(c)

B1: Any valid reason. See scheme for alternatives

Q12.

Question	Scheme	Marks	AOs
	(a) $f(x) = -3x^3 + 8x^2 - 9x + 10, x \in \mathbb{R}$		
(a)	(i) $\{f(2) = -24 + 32 - 18 + 10 \Rightarrow\} f(2) = 0$	B1	1.1b
	(ii) $\{f(x) = (x-2)(-3x^2 + 2x - 5) \text{ or } (2-x)(3x^2 - 2x + 5)\}$	M1	2.2a
		A1	1.1b
		(3)	
(b)	$-3y^6 + 8y^4 - 9y^2 + 10 = 0 \Rightarrow (y^2 - 2)(-3y^4 + 2y^2 - 5) = 0$		
	Gives a partial explanation by - explaining that $-3y^4 + 2y^2 - 5 = 0$ has no {real} solutions with a reason, e.g. $b^2 - 4ac = (2)^2 - 4(-3)(-5) = -56 < 0$ - or stating that $y^2 = 2$ has 2 {real} solutions or $y = \pm\sqrt{2}$ {only}	M1	2.4
	Complete proof that the given equation has exactly two {real} solutions	A1	2.1
		(2)	
(c)	$3 \tan^3 \theta - 8 \tan^2 \theta + 9 \tan \theta - 10 = 0; 7\pi \leq \theta < 10\pi$		
	{Deduces that} there are 3 solutions	B1	2.2a
		(1)	
(6 marks)			

Notes for Question	
(a)(i)	
B1:	$f(2) = 0$ or 0 stated by itself in part (a)(i)
(a)(ii)	
M1:	Deduces that $(x-2)$ or $(2-x)$ is a factor and attempts to find the other quadratic factor by - using long division to obtain either $\pm 3x^2 \pm kx + \dots, k = \text{value} \neq 0$ or $\pm 3x^2 \pm \alpha x + \beta, \beta = \text{value} \neq 0, \alpha$ can be 0 - factorising to obtain their quadratic factor in the form $(\pm 3x^2 \pm kx \pm c), k = \text{value} \neq 0, c$ can be 0, or in the form $(\pm 3x^2 \pm \alpha x \pm \beta), \beta = \text{value} \neq 0, \alpha$ can be 0
A1:	$(x-2)(-3x^2 + 2x - 5), (2-x)(3x^2 - 2x + 5)$ or $-(x-2)(3x^2 - 2x + 5)$ stated together as a product
(b)	
M1:	See scheme
A1:	See scheme. Proof must be correct <i>with no errors</i> , e.g. giving an incorrect discriminant value
Note:	Correct calculation e.g. $(2)^2 - 4(-3)(-5), 4 - 60$ or -56 must be given for the first explanation
Note:	Note that M1 can be allowed for - a correct follow through calculation for the discriminant of their "$-3y^4 + 2y^2 - 5$" which would lead to a value < 0 together with an explanation that $-3y^4 + 2y^2 - 5 = 0$ has no {real} solutions - or for the omission of < 0
Note:	< 0 must also be stated in a discriminant method for A1
Note:	Do not allow A1 for incorrect working, e.g. $(2)^2 - 4(-3)(-5) = -54 < 0$
Note:	$y^2 = 2 \Rightarrow y = \pm 2$, so 2 solutions is not allowed for A1, but can be condoned for M1
Note:	Using the formula on $-3y^4 + 2y^2 - 5 = 0$ or $-3x^2 + 2x - 5 = 0$ gives y^2 or $x = \frac{-2 \pm \sqrt{-56}}{-6}$ or $\frac{-1 \pm \sqrt{-14}}{-3}$
Note:	Completing the square on $-3x^2 + 2x - 5 = 0$ gives $x^2 - \frac{2}{3}x + \frac{5}{3} = 0 \Rightarrow \left(x - \frac{1}{3}\right)^2 - \frac{1}{9} + \frac{5}{3} = 0 \Rightarrow x = \frac{1}{3} \pm \sqrt{\frac{-14}{9}}$
Note:	Do not recover work for part (b) in part (c)
(c)	
B1:	See scheme
Note:	Give B0 for stating $\theta = \text{awrt } 23.1, \text{awrt } 26.2, \text{awrt } 29.4$ without reference to 3 solutions